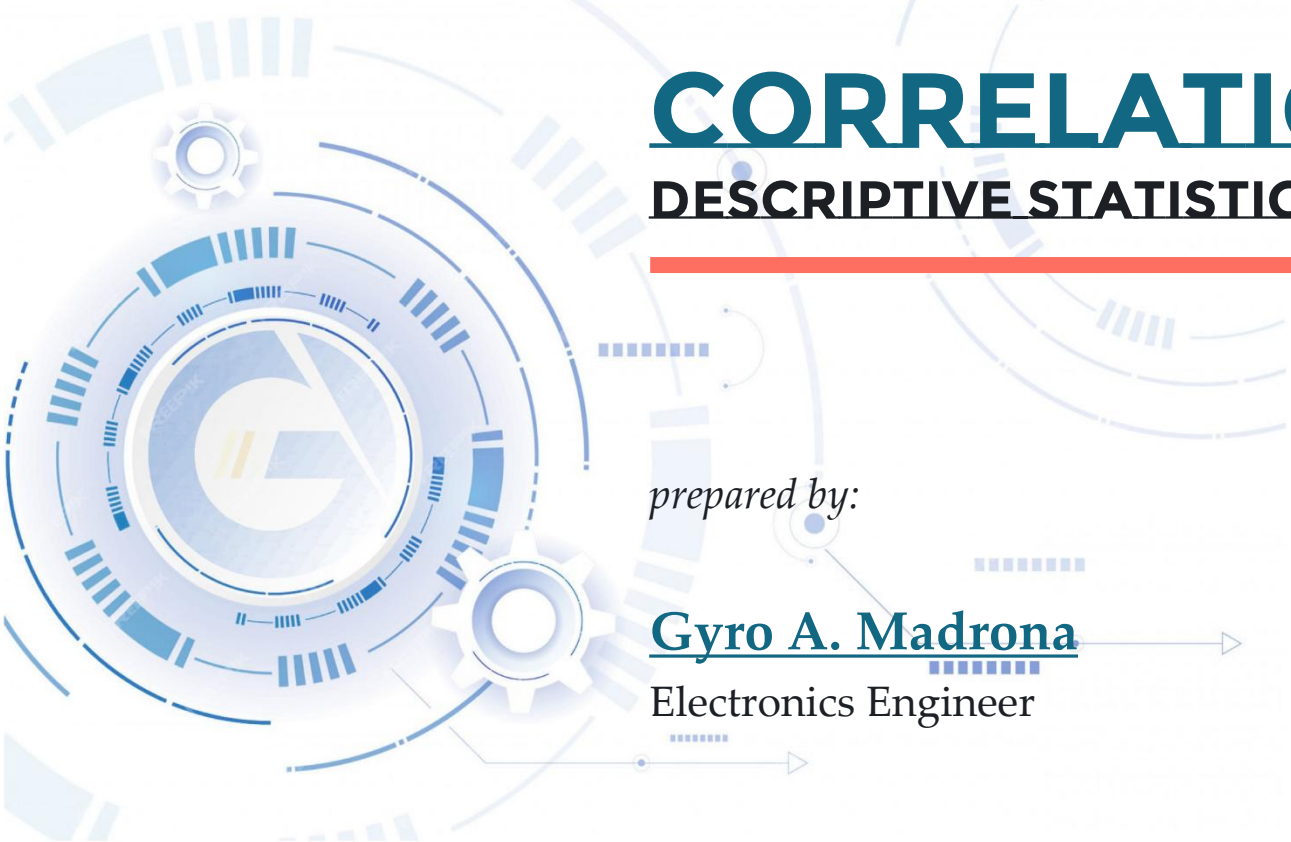




CORRELATION

DESCRIPTIVE STATISTICS

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Covariance

Correlation Coefficient



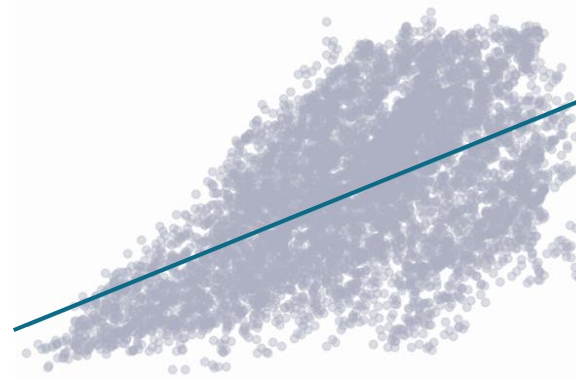
COVARIANCE



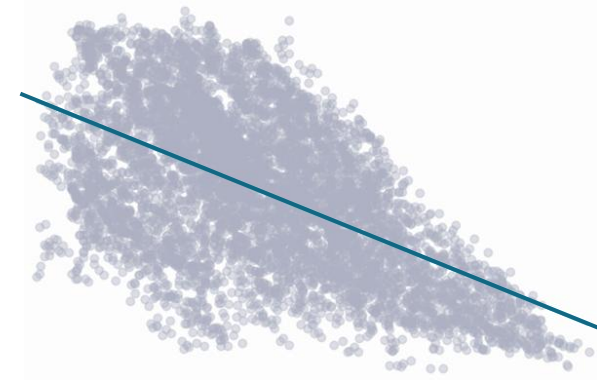
COVARIANCE

Covariance is a statistical measure that quantifies the relationship between two random variables (X, Y).

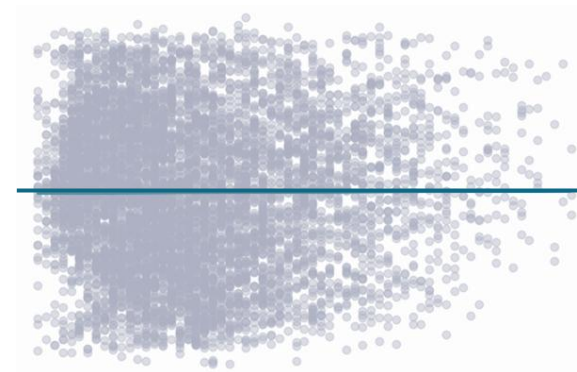
Scatter Plot



Covariance > 0



Covariance < 0



Covariance $= 0$



COVARIANCE

Covariance is a statistical measure that quantifies the relationship between two random variables (X, Y).

Population Covariance

$$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



COVARIANCE

Covariance is a statistical measure that quantifies the relationship between two random variables (X, Y).

The `df.cov()` method is used to compute the covariance matrix of a DataFrame.



EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
3.5	200
1.8	90
4.2	210
2.7	110

$$\bar{x} = 2.84$$

Solution

let $x = \text{current}$

$y = \text{power}$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{x} = \frac{2 + 3.5 + 1.8 + 4.2 + 2.7}{5}$$

$$\bar{x} = 2.84$$



EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
3.5	200
1.8	90
4.2	210
2.7	110

$$\bar{x} = 2.8A \quad \bar{y} = 142$$

Solution

let $x = \text{current}$
 $y = \text{power}$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$\bar{y} = \frac{100 + 200 + 90 + 210 + 110}{5}$$

$$\bar{y} = 142$$



EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
3.5	200
1.8	90
4.2	210
2.7	110

$$\bar{x} = 2.84 \quad \bar{y} = 142$$

Solution

Covariance

$$S_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$S_{xy} = \frac{(2-2.84)(100-142) + \dots + (2.7-2.84)(110-142)}{5-1}$$

$$S_{xy} = 56.15$$

ans

This indicates positive linear relationship.
As current increases, power consumption tends to increase.

But how **strong** is the relationship?

CORRELATION COEFFICIENT



CORRELATION COEFFICIENT

Correlation coefficient adjusts covariance, so that the relationship between the two variables becomes easy and intuitive to interpret.

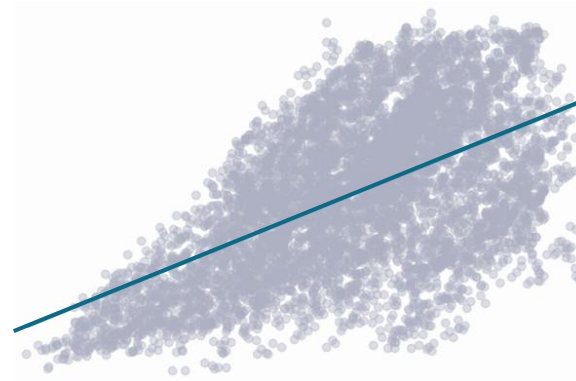
It ranges from -1 to $+1$:

$+1$ indicates perfect positive correlation

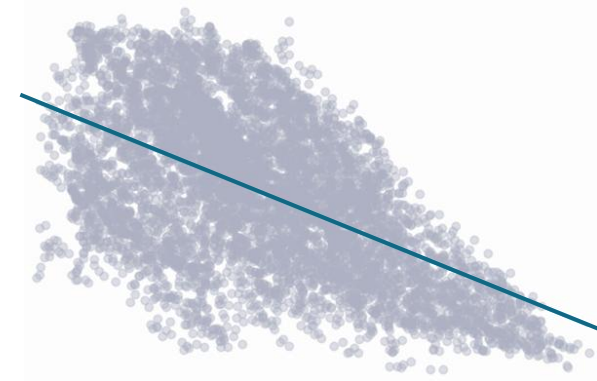
-1 indicates perfect negative correlation

0 indicates no linear relationship

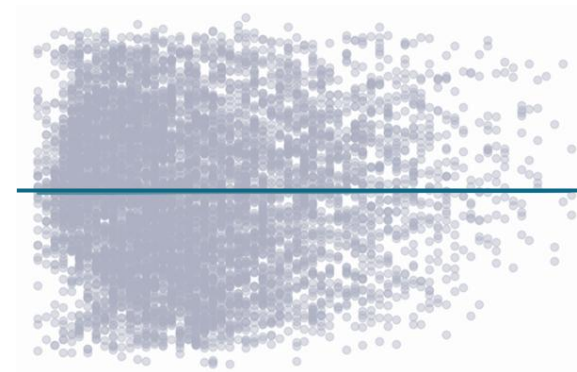
Scatter Plot



Correlation > 0



Correlation < 0



Correlation $= 0$



CORRELATION COEFFICIENT

Correlation coefficient adjusts covariance, so that the relationship between the two variables becomes easy and intuitive to interpret.

It ranges from -1 to $+1$:

$+1$ indicates perfect positive correlation

-1 indicates perfect negative correlation

0 indicates no linear relationship

Population Correlation Coefficient

$$r = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Sample Correlation Coefficient

$$r = \frac{s_{xy}}{s_x s_y}$$

Pearson's correlation coefficient



CORRELATION COEFFICIENT

Correlation coefficient adjusts covariance, so that the relationship between the two variables becomes easy and intuitive to interpret.

It ranges from -1 to $+1$:

$+1$ indicates perfect positive correlation

-1 indicates perfect negative correlation

0 indicates no linear relationship

The `df.corr()` method is used to compute the correlation matrix of a DataFrame.



EXERCISE

Determine if each scenario suggests a positive, negative, or no correlation:

1. Ice cream sales and umbrella sales in a city. → negative correlation
2. Hours spent studying and exam scores. → positive correlation
3. A person's shoe size and their IQ. → no correlation
4. Age of a used car and its resale value. → negative correlation



EXERCISE

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Device	
Current	Power
2	100
3.5	200
1.8	90
4.2	210
2.7	110

$$\bar{x} = 2.84 \quad \bar{y} = 142 \quad S_{xy} = 56.15$$

$$S_x = 1.011$$

Solution

$$S_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$S_x^2 = \frac{(2-2.84)^2 + \dots + (2.7-2.84)^2}{5-1}$$

$$S_x^2 = 1.023$$

$$\underline{S_x = 1.011}$$



EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
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2.7	110

$$\bar{x} = 2.84 \quad \bar{y} = 142 \quad s_{xy} = 56.15$$

$$s_x = 1.011 \quad s_y = 58.15$$

Solution

$$s_y^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$$

$$s_y^2 = \frac{(100-142)^2 + \dots + (110-142)^2}{5-1}$$

$$s_y^2 = 3370$$

$$\underline{s_y = 58.05}$$



EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
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2.7	110

$$\bar{x} = 2.84 \quad \bar{y} = 142 \quad S_{xy} = 56.15$$

$$S_x = 1.011 \quad S_y = 58.15$$

Solution

Correlation coefficient

$$r = \frac{S_{xy}}{S_x S_y}$$

$$r = \frac{56.15}{(1.011)(58.15)}$$

$$r = 0.957$$

ans

x + 1

this indicates **strong** positive linear relationship. As current increases, power consumption tends to increase.

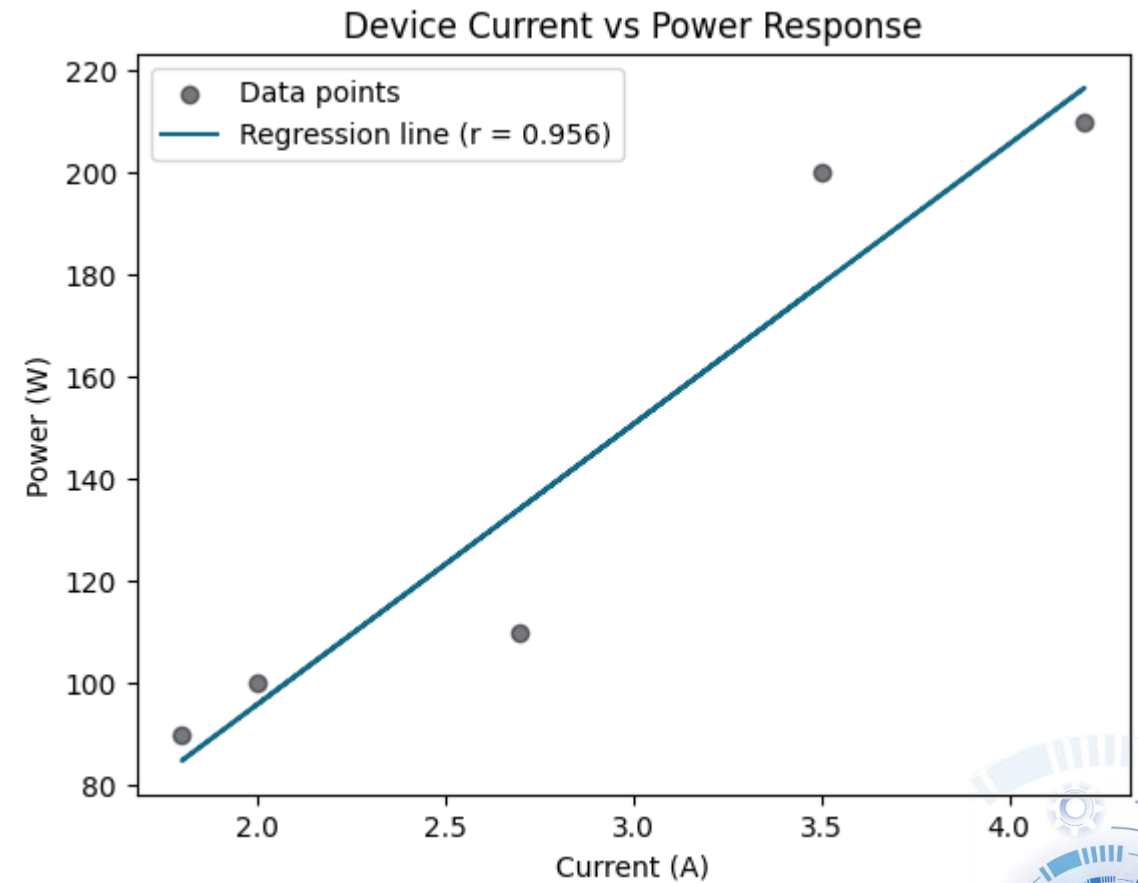


EXERCISE

The given dataset contains five observations of current (A) and corresponding power (W) measurements. Does **current** and **power** consumption have a positive, negative, or no linear relationship?

Device	
Current	Power
2	100
3.5	200
1.8	90
4.2	210
2.7	110

Solution



LABORATORY

